

	1	2	3	4	5	6	7	8
A	Integrated draft: power chain (sheets 2–4) and board stage (sheets 5–8). See DESIGN_SPEC.md. Rails: V5_SYS from the 5 V inputs, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. No PCB placement or routing has been performed.							
B	POWER 1 / 5 V in, USB, priority power mux File: power_sources.kicad_sch				BOARD 2 / IMU, barometer, high-g File: sensors.kicad_sch			
C	POWER 2 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch				BOARD 3 / log flash and microSD File: storage.kicad_sch			
D	POWER 3 / peripheral ports File: power_periph.kicad_sch				BOARD 4 / PWM outputs File: actuators.kicad_sch			
E	BOARD 1 / RP2354B File: mcu.kicad_sch							
F	1	2	3	4	5	6	7	8

Sheet: / File: MARV-V2.kicad_sch		
Title: MARV integrated power – preliminary		
Size: A3	Date: 2026-09-14	Rev: P0 – NOT FOR FAB
KiCad E.D.A. 10.0.6	Id: 1/8	

5 V INPUTS ONLY: BEC/boost 4.75–5.5 V (J3), USB VBUS ~5 V (J4). NO RAW CELL INPUT ON THIS BOARD.

A

B

C

D

E

F

A

B

C

D

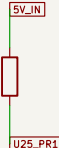
E

F

D23
SMAJ5.0A, 5.0 V standoff uni-directional TVS



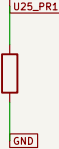
R42
32.4k / 1%, PR1 top (5V_IN)



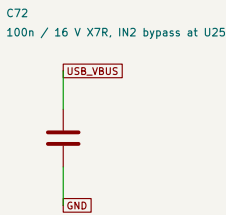
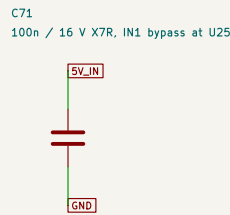
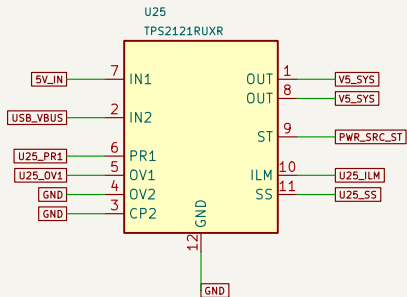
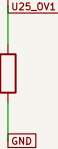
R44
45.3k / 1%, OV1 top (5V_IN)



R43
10k / 1%, PR1 bottom



R45
10k / 1%, OV1 bottom



#FLG1
PWR_FLAG



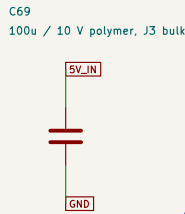
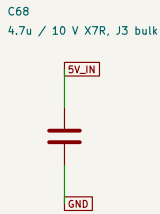
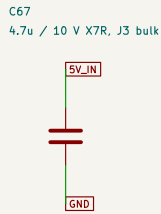
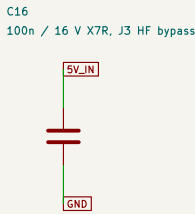
#FLG2
PWR_FLAG



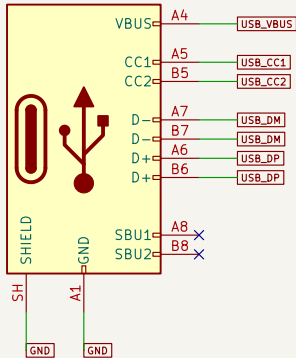
#FLG3
PWR_FLAG



#FLG4
PWR_FLAG



J4
USB_C_PROGRAM_POWER



R4
5.1k / 1%



R5
5.1k / 1%



U25 TPS2121 PRIORITY POWER MUX (TI SLVSEA3F). IN1 = 5V_IN is the priority source, IN2 = USB_VBUS the fallback; OUT = V5_SYS. Reverse-current blocking is always on for BOTH channels (Sec 9.3.6, IRCB 0.2/1/2 A, tRCB 10 us, VRCB 0/25/50 mV), so neither input can be back-fed from V5_SYS or from the other input. RON 56 mOhm typ / 100 mOhm max over temperature (Sec 7.5, VINx >= 5 V). PR1 (Sec 10.2.4.1 Eq.5, VPR1 = VIN1 x Rb/(Rt+Rb) compared with VREF): R42 32.4k / R43 10k -> IN1 released when 5V_IN falls to 1.04 V / 0.23585 = 4.410 V typ and re-selected at 1.06 V / 0.23585 = 4.494 V typ (4.20–4.62 V / 4.28–4.66 V over the VREF spec). The divider is centred on the RISING edge, not the falling one: the binding corner is a 5V_IN at its 4.75 V minimum meeting a VREF at its 1.10 V maximum, and 4.664 V worst-case rising keeps IN1 selected everywhere inside the declared 4.75–5.5 V window. OV1 (Sec 9.3.5 / 10.2.4.2 Eq.6, same divider form): R44 45.3k / R45 10k -> IN1 rejected at 1.06 V / 0.180832 = 5.862 V typ (5.585–6.083 V over the VREF spec, so it can never trip inside the declared 4.75–5.5 V input window). OV2 is tied to GND: the USB channel has no overvoltage cut-off ("Connect to GND if not required", Sec 6). CP2 is tied to GND, which selects the internal-VREF priority scheme of Table 9–3 – that is the mode PR1–vs–VREF describes, and it costs the TPS2121 fast-switchover path: the applicable spec is tSW = 100 us typ (Sec 7.5), not the 5 us tFSW that needs CP2 >= VREF. ILM (Sec 9.3.2 Eq.2, ILM = 65.2 / RILM'0.861 with RILM in kOhm, valid 18–100 kOhm): R46 80.6k -> 1.49 A typ; the datasheet characterises RILM = 80k as 1.0 / 1.5 / 2.0 A min/typ/max. Fast-trip OCP is 2.4 x ILM (Sec 9.3.3). SS (Sec 9.3.1 / Table 9–1): C70 100 nF -> 780 V/s at 5 V, so the ~240 uF on V5_SYS draws ~190 mA of inrush and OUT ramps in ~6.4 ms. ST (Sec 6, Sec 10.2.3) is an open-drain status output; HIGH when IN1 (or neither input) drives OUT, LOW when IN2 does, R47 10k to V3V3_SYS is inside the RST = 6–20 kOhm recommended operating range of Sec 7.3; PWR_SRC_ST lands on RP2354B GPIO39 (pin 48). Sec 11 gives no numeric minimum COUT, only "increase the capacitance on OUT to avoid output voltage drop"; C19 220 uF + C8/C9 2 x 10 uF on POWER 2 already exceed the 100–200 uF the datasheet design examples use, so no extra output capacitor is added here.

J3 INPUT NETWORK: C16 100 nF + C67/C68 2 x 4.7 uF X7R + C69 100 uF polymer, and D23 SMAJ5.0A (DO–214AC/SMA) clamping 5V_IN to GND. D23 is a UNI-directional TVS: its KiCad symbol inherits the generic A1/A2 pin names from the SM6T family, but the graphic and the DO–214AC band both put the CATHODE on pin 1, so pin 1 goes to 5V_IN and pin 2 (anode) to GND. 5.0 V standoff / 6.4 V VBR min sits above the 5.5 V input maximum and below the TPS2121 5.585 V worst-case OV1 trip, so the clamp and the mux do not fight. J3 is XT30PW–M: pad 1 is the "–" terminal and pad 2 the "+" terminal in the KiCad footprint – see the comment in tools/build_power.py.

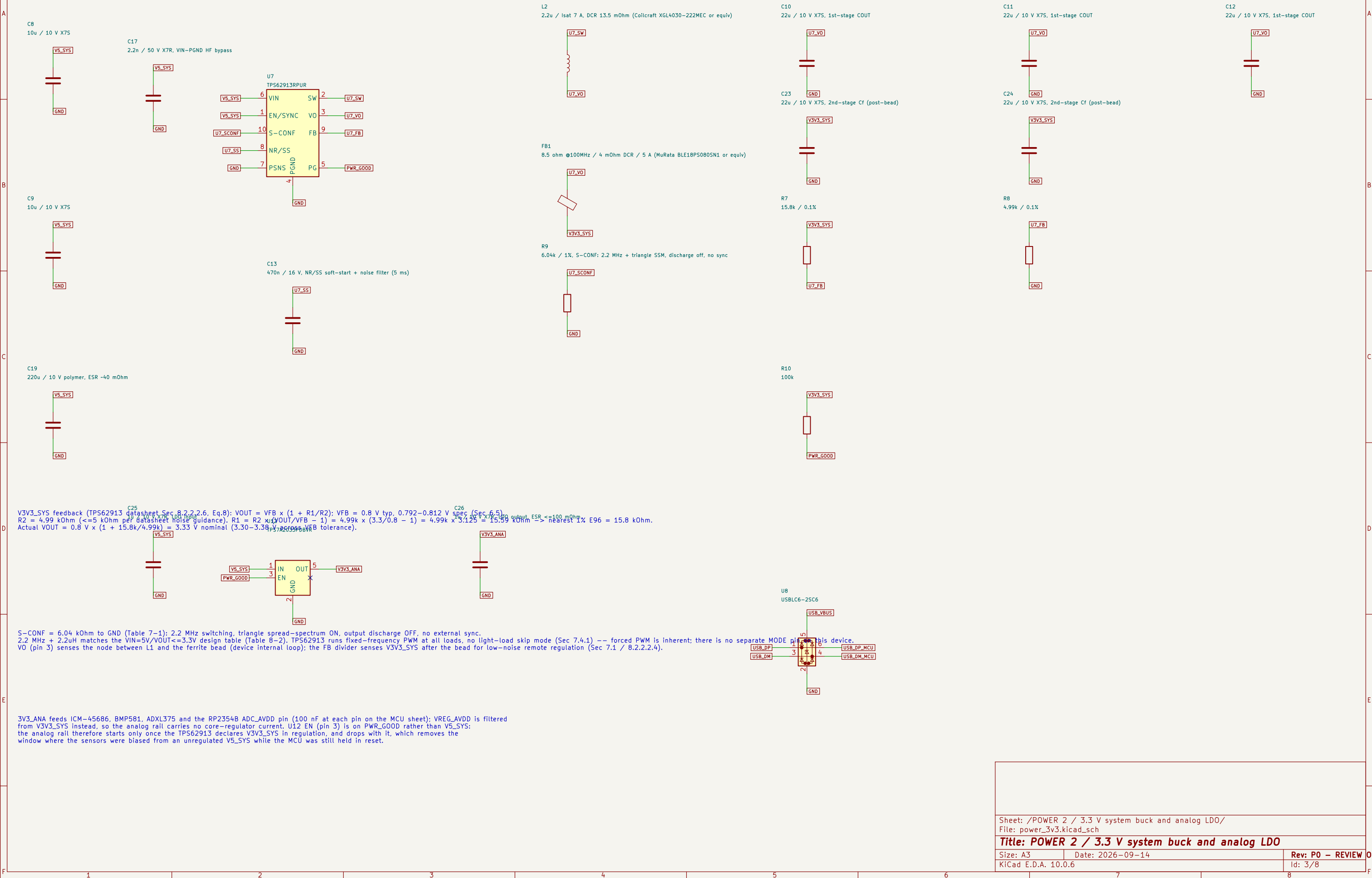
USB_VBUS feeds U25 IN2 directly; no inrush/current limiting on this sheet.
Data ESD protection is on POWER 2. No servo rail connection.

C7
1u / 10 V

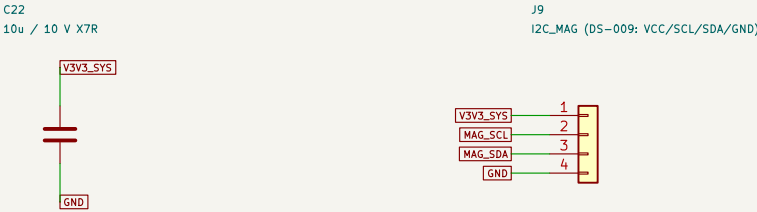
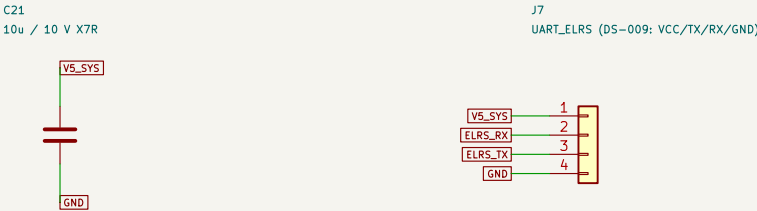
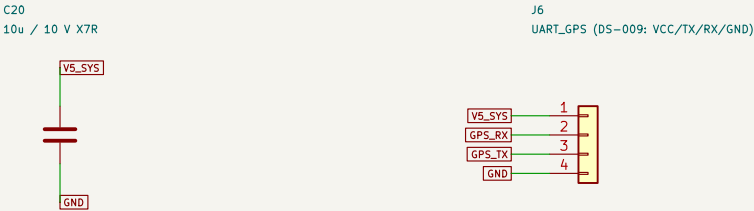


Sheet: /POWER 1 / 5 V in, USB, priority power mux/ File: power_sources.kicad_sch		
Title: POWER 1 / 5 V in, USB, priority power mux		
Size: A3	Date: 2026–09–14	Rev: P0 – REVIEW ONLY
KiCad E.D.A. 10.0.6		Id: 2/8

AVIONICS ONLY: 300 mA continuous / 500 mA short peak, provisional. NO SERVO POWER.

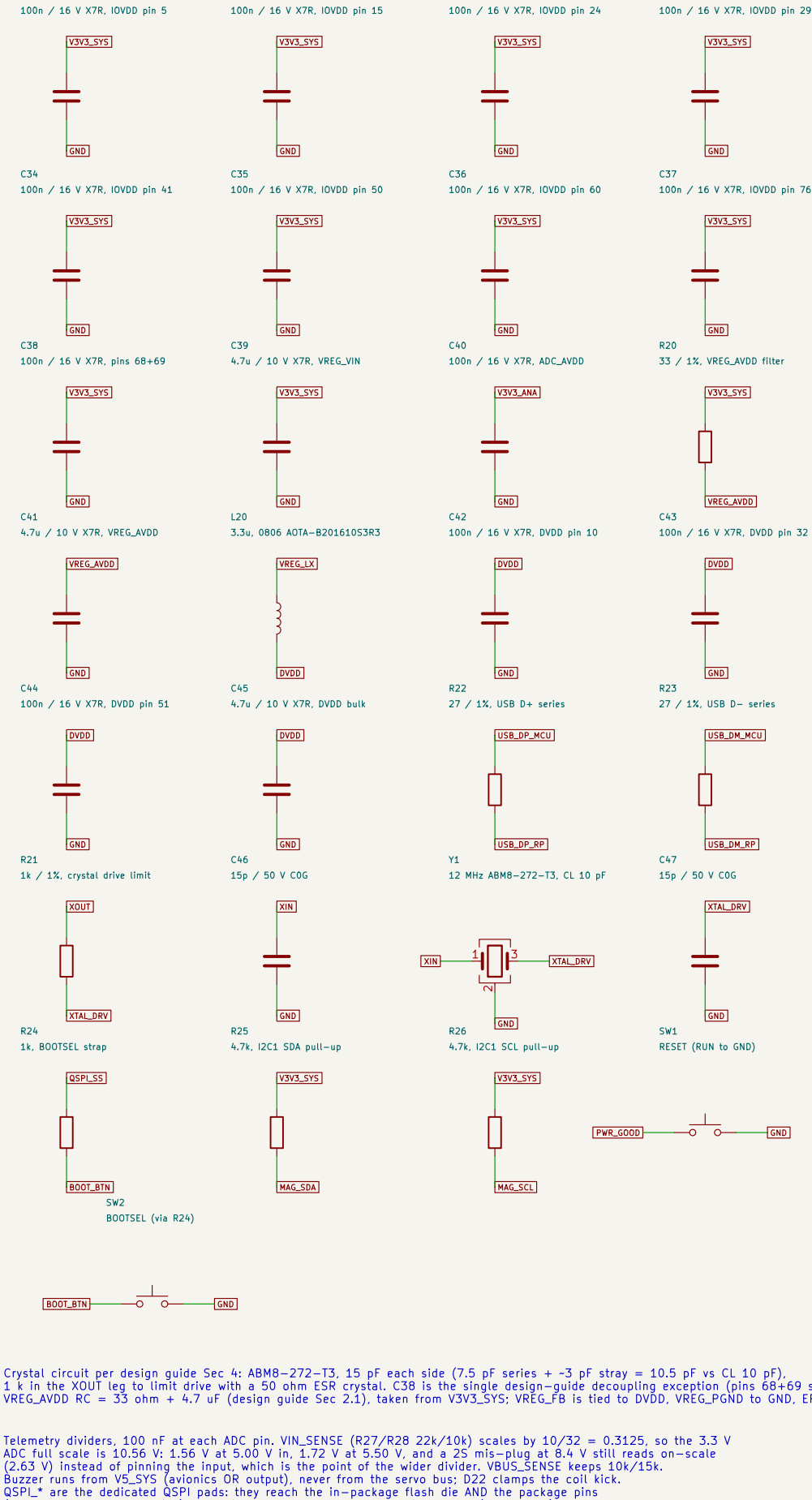


PIXHAWK DS-009 PIN ORDER. UART ports (J6, J7): 1 VCC, 2 TX, 3 RX, 4 GND. I2C port (J9): 1 VCC, 2 SCL, 3 SDA, 4 GND. Pin 2 of a UART port is the BOARD TX, i.e. the RP2354B UART TX output, which lands on the module RX input – that is the GPS_RX / ELRS_RX net, because a port label is named for the peripheral pin it belongs to (see the MCU sheet). Pin 3 is the board RX, fed by the module TX: GPS_TX / ELRS_TX.



RAILS: J6 and J7 VCC are on V5_SYS – GNSS and ELRS modules are 5 V–powered with their own on–module regulators, and their C20 / C21 10 uF local bulk moved to V5_SYS with them. Their SIGNAL pins stay 3.3 V CMOS straight from the RP2354B, which is what both module families expect. J9 VCC stays on V3V3_SYS: the magnetometer bus is 3.3 V and its pull–ups (R25/R26 on the MCU sheet) bias to V3V3_SYS, so a 5 V port pin there would fight them. J9 pin 2 = MAG_SCL, pin 3 = MAG_SDA.

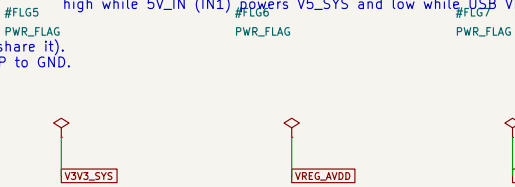
RP2354B, QFN-80, 2 MB in-package QSPI flash. Digital rails on V3V3_SYS, core DVDD from the on-chip buck.
RAIL SPLIT: V3V3_ANA (TPS7A20 LDO) now feeds exactly one RP2354B pin – ADC_AVDD (pin 59, with C40). The core regulator input filter R20/C41 that produces VREG_AVDD is fed from V3V3_SYS instead, so the buck-converter reference current no longer flows out of the low-noise analog rail and the sensors do not share a rail with it; V3V3_ANA is left carrying only ADC_AVDD and the MEMS. Decoupling, core regulator, crystal and BOOTSEL follow the RP2350 hardware design guide; every pin net is a global label.



Crystal circuit per design guide Sec 4: ABM8-272-T3, 15 pF each side (7.5 pF series + ~3 pF stray = 10.5 pF vs CL 10 pF), 1 k in the XOUT leg to limit drive with a 50 ohm ESR crystal. C38 is the single design-guide decoupling exception (pins 68+69 share it). VREG_AVDD RC = 33 ohm + 4.7 uF (design guide Sec 2.1), taken from V3V3_SYS; VREG_FB is tied to DVDD, VREG_PGND to GND, EP to GND.

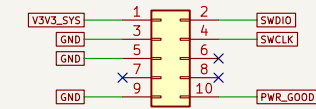
Telemetry dividers, 100 nF at each ADC pin. VIN_SENSE (R27/R28 22k/10k) scales by 10/32 = 0.3125, so the 3.3 V ADC full scale is 10.56 V: 1.56 V at 5.00 V in, 1.72 V at 5.50 V, and a 2S mis-plug at 8.4 V still reads on-scale (2.63 V) instead of pinning the input, which is the point of the wider divider. VBUS_SENSE keeps 10k/15k. Buzzer runs from V5_SYS (avionics OR output), never from the servo bus; D22 clamps the coil kick. QSPI_* are the dedicated QSPI pads: they reach the in-package flash die AND the package pins (RP2350 datasheet Sec 14.3), so U24 shares the bus with GPIO0/QMI CS1n (FLASH_CS1) as its select. RP2354 requires QSPI_IOVDD = 3.3 V, and IOVDD = 3.3 V to run a second QSPI device (Sec 14.9).

PORT NAMING: a port label is named for the peripheral pin it comes from, so GPS_TX / ELRS_TX are module outputs and land on MCU UART RX pins, while GPS_RX / ELRS_RX are module inputs fed by the MCU UART TX pins. On J6/J7 POWER 3, Pixhawk DS-009 order 1 VCC / 2 TX / 3 RX / 4 GND) pin 2 is therefore GPS_RX / ELRS_RX. J6 and J7 VCC are on V5_SYS; J9 VCC stays on V3V3_SYS. All port SIGNAL pins remain 3.3 V CMOS. GPIO39 (pin 48) is no longer spare: it reads PWR_SRC_ST, the open-drain ST output of U25 (TPS2121), which is high while 5V_IN (IN1) powers V5_SYS and low while USB VBUS (IN2) does.

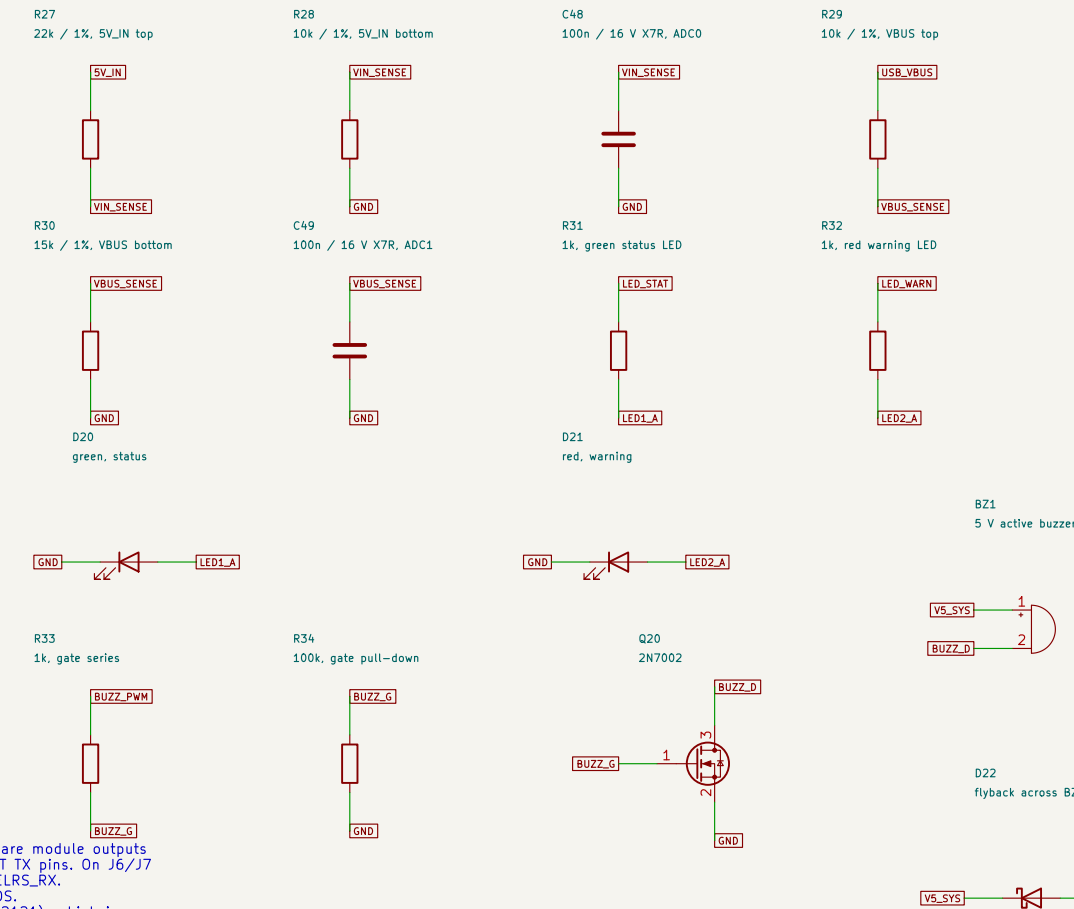


GPIO MAP (pin = QFN-80 pin)		
GPIO0	77	FLASH_CS1
GPIO1	78	(no connect) spare
GPIO2	79	PWM1
GPIO3	80	PWM2
GPIO4	1	PWM3
GPIO5	2	PWM4
GPIO6	3	MAG_SDA
GPIO7	4	MAG_SCL
GPIO8	6	ELRS_RX
GPIO9	7	ELRS_TX
GPIO10	8	PWM5
GPIO11	9	PWM6
GPIO12	11	GPS_RX
GPIO13	12	GPS_TX
GPIO14	13	PWM7
GPIO15	14	PWM8
GPIO16	16	SENS_MISO
GPIO17	17	HG_ACC_CS
GPIO18	18	SENS_SCK
GPIO19	19	SENS_MOSI
GPIO20	20	IMU_CS
GPIO21	21	(no connect) spare
GPIO22	22	BARO_CS
GPIO23	23	IMU_INT1
GPIO24	25	IMU_INT2
GPIO25	26	BARO_INT
GPIO26	27	LED_STAT
GPIO27	28	LED_WARN
GPIO28	36	BUZZ_PWM
GPIO29	37	HG_ACC_INT
GPIO30	38	(no connect) spare
GPIO31	39	(no connect) spare
GPIO32	40	SD_CLK
GPIO33	42	SD_CMD
GPIO34	43	SD_D0
GPIO35	44	SD_D1
GPIO36	45	SD_D2
GPIO37	46	SD_D3
GPIO38	47	SD_DET
GPIO39	48	PWR_SRC_ST
GPIO40	49	VIN_SENSE
GPIO41	52	VBUS_SENSE
GPIO42	53	(no connect) ADC2 spare
GPIO43	54	(no connect) ADC3 spare
GPIO44	55	(no connect) ADC4 spare
GPIO45	56	(no connect) ADC5 spare
GPIO46	57	(no connect) ADC6 spare
GPIO47	58	(no connect) ADC7 spare

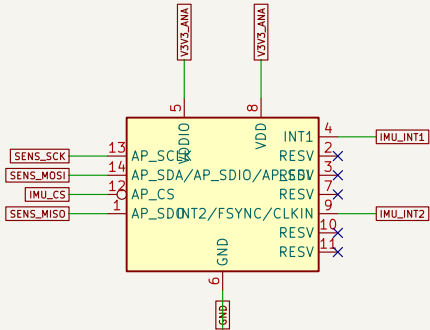
J10
SWD, Cortex-M 10-pin 1.27 mm



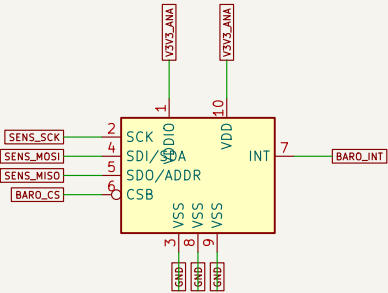
J10 pin 1 is 3V3 OUT for debugger reference only – never power the board from it. Pin 10 is the debugger reset line on PWR_GOOD/RUN: pins 6-8 (SWD, KEY, NC) unused. RUN is held by the TPS62913 power-good pull-up on POWER 2, so the MCU cannot run before V3V3_SYS is in regulation.



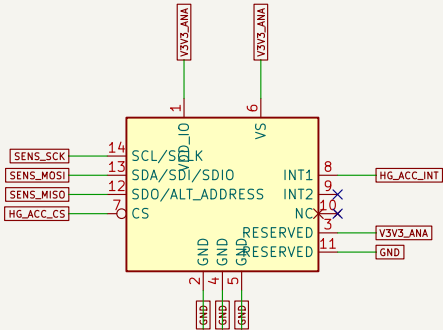
All three sensors run 4-wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TPS7A20 LDO). Every supply pin on this sheet gets 100 nF + 1 uF; BMP581 and ADXL375 because their own datasheets call for the pair, the ICM-45686 as a deliberate addition – its datasheet BOM (DS-000577 Rev 1.0 Table 11) lists 0.1 uF only.



U21
ICM-45686 6-axis IMU



U22
BMP581 barometer



U23
ADXL375 high-g accelerometer

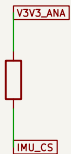
C50
100n / 16 V X7R, U21 VDD



C51
1u / 10 V X7R, U21 VDD



R48
10k, IMU_CS pull-up



C54
100n / 16 V X7R, U22 VDD



C55
1u / 10 V X7R, U22 VDD



C58
100n / 16 V X7R, U23 VS



C59
1u / 10 V X7R, U23 VS



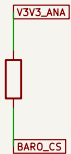
C52
100n / 16 V X7R, U21 VDDIO



C53
1u / 10 V X7R, U21 VDDIO



R50
10k, BARO_CS pull-up



R51
10k, HG_ACC_CS pull-up



C56
100n / 16 V X7R, U22 VDDIO



C57
1u / 10 V X7R, U22 VDDIO



C60
100n / 16 V X7R, U23 VDD_IO



C61
1u / 10 V X7R, U23 VDD_IO



U21 ICM-45686 (TDK InvenSense DS-000577 Rev 1.0, Table 10 "Signal Descriptions" / Figure 4 pin-out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = IMU_CS, AP_SCLK (13) = SENS_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = SENS_MOSI, AP_SDO/AP_ADO (1) = SENS_MISO, INT1 (4) = IMU_INT1, INT2/FSYNC/CLKIN (9) = IMU_INT2 – typed bidirectional because FSYNC/CLKIN are host-driven inputs while INT2 is a chip output; only the INT2 function is used here, VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single-Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per-pin notes in Table 10); they are only needed for the AUX1 or I2C-master dual-interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 "Bill of Materials" (C1/C2 in the datasheet); C51 (VDD) and C53 (VDDIO) add 1 uF on top of them. The datasheet BOM lists 0.1 uF only; this board keeps 100 nF + 1 uF per supply pin for consistency with U22/U23 and common flight-controller practice.

U22 BMP581: CSB = BARO_CS, SDI = SENS_MOSI, SDO = SENS_MISO, SCK = SENS_SCK, INT = BARO_INT, VDDIO and VDD on V3V3_ANA, all three VSS pins to GND.
U23 ADXL375: CS = HG_ACC_CS, SDA/SDI = SENS_MOSI, SDO = SENS_MISO, SCL/SCLK = SENS_SCK, INT1 = HG_ACC_INT. Per the pin function table, pin 3 (RESERVED) goes to VS and pin 11 (RESERVED) to GND; pin 10 (NC) and the unused INT2 are left open. VS and VDD I/O are both on V3V3_ANA.

R48, R50, R51: 10k pull-ups to V3V3_ANA on the three SPI chip selects (IMU_CS, BARO_CS, HG_ACC_CS). R49 (the former second IMU chip-select pull-up) is removed along with that net – the ICM-45686 has a single chip select. Same rationale as R35 on FLASH_CS1 (BOARD_3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull-up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

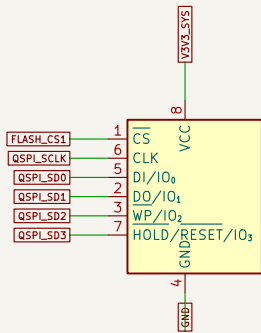
Sheet: /BOARD 2 / IMU, barometer, high-g/
File: sensors.kicad_sch

Title: BOARD 2 / IMU, barometer, high-g

Size: A3
Date: 2026-09-14
KiCad E.D.A. 10.0.6

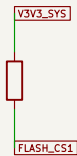
Rev: P0 – REVIEW ONLY
Id: 6/8

Log flash on the shared QSPI bus with GPIO0/QMI CS1n as its chip select, and a latched microSD wired for 4-bit PIO (also SPI-compatible). Both on V3V3_SYS with local bulk, per the rail split in DESIGN_SPEC.



U24
W25Q128JVSIQ, 128 Mbit QSPI NOR

R35
10k, FLASH_CS1 pull-up



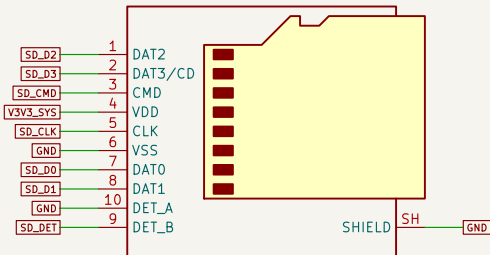
C62
100n / 16 V X7R, U24 VCC



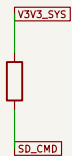
C63
1u / 10 V X7R, U24 VCC



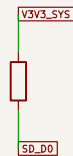
J11
microSD, Molex 104031-0811 push-push



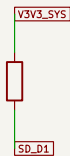
R36
10k, SD_CMD pull-up



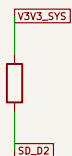
R37
10k, SD_D0 pull-up



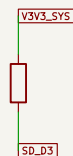
R38
10k, SD_D1 pull-up



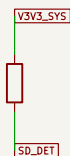
R39
10k, SD_D2 pull-up



R40
10k, SD_D3 pull-up



R41
10k, SD_DET pull-up



C64
100n / 16 V X7R, at the socket



C65
10u / 10 V X7R, at the socket



C66
47u / 6.3 V X5R, at the socket



U24 shares the dedicated QSPI pads with the RP2354B in-package flash die (datasheet Sec 14.3): CLK = QSPLSCLK, D1/IO0 = QSPLSD0, D0/IO1 = QSPLSD1, WP/IO2 = QSPLSD2, HOLD/IO3 = QSPLSD3. Only the chip select differs – FLASH_CS1 comes from GPIO0 (QMI CS1n), pulled up to V3V3_SYS by R35 so the part is deselected before the MCU drives it. QSPI_SS (the internal die) keeps its own strap on the MCU sheet.

J11: DET_A to GND and DET_B to SD_DET with a 10k pull-up, so the input reads low with a card inserted. The shield goes to GND. 100 nF + 10 uF + 47 uF sit at the socket (DESIGN_SPEC "Rails": SD transients stay on the buck rail), 10k pull-ups on CMD and DAT0-DAT3 are the SD-standard idle bias and also keep DAT3/CD high for 4-bit mode. VDD = V3V3_SYS.

Sheet: /BOARD 3 / log flash and microSD/
File: storage.kicad_sch

Title: BOARD 3 / log flash and microSD

Size: A3
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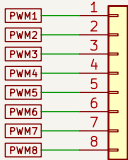
Date: 2026-09-14

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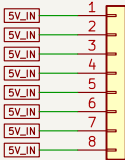
Id: 7/8

Eight PWM outputs on a 3-row 2.54 mm block: signal /2SERVO_5V / GND. No series resistors –3.3 V logic straight from the RP2354B, which is what servos and ESCs expect.

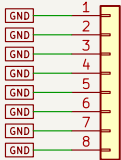
J12
PWM signals, row 1



J13
SERVO_5V pass-through, row 2



J14
GND, row 3



3.3 V logic; servo/ESC power is the external BEC bus, 3 A limit is the BEC’s.

Row 2 is SERVO_5V = the 5V_IN net, i.e. the J3 input BEFORE U25 (the TPS2121 priority power mux). Servo and ESC current therefore never crosses the board rails, V5_SYS or the mux; in particular it is not counted against the mux ILM current limit (1.49 A, set by R46). It returns through the row 3 ground pins. Nothing on this sheet loads V3V3_SYS or USB_VBUS.

No 3-row generic connector symbol or 2.54 mm 3x08 footprint ships with this KiCad install, so the header is drawn as three stacked Conn_01x08 symbols (J12 signal, J13 SERVO_5V, J14 GND) with three 1x08 footprints. Place them on 2.54 mm centres to form the usual 3x8 servo block; column n is J12 pin n / J13 pin n / J14 pin n.

PWM MAP
PWM1 = GPIO2 PWM5 = GPIO10
PWM2 = GPIO3 PWM6 = GPIO11
PWM3 = GPIO4 PWM7 = GPIO14
PWM4 = GPIO5 PWM8 = GPIO15

Sheet: /BOARD 4 / PWM outputs/
File: actuators.kicad_sch

Title: BOARD 4 / PWM outputs

Size: A3 Date: 2026-09-14
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